$A+(B+C)=$

$x+0=x=0+x$

MTH405 lecture 1 to 22 midterm mcqs

(1) $(Z,-)$ is a groupoid under the operation of usual Substraction. (True)

(2) Length of cyclic permutation is $\begin{pmatrix}1&2&3&4&5\\1&2&4&5&3\end{pmatrix}$
= Same elements ignore
$= (3\ 4\ 5) = 3$

(3) Length of cyclic ~~group~~ permutation $\begin{pmatrix}d_1&d_2&d_3\\d_2&d_3&d_1\end{pmatrix}$
$= 3$

(4) Multiplication is a binary operation on the Set of Natural number N. (True)

(5) Length of cyclic permutation $\begin{pmatrix}2&3&1&4&5&6\\1&3&5&4&2&6\end{pmatrix}$
= Same element ignore
$= \cancel{2\ 1\ 5}$ $\begin{pmatrix}2&1&5\\1&5&2\end{pmatrix} = 3$

(6) A non empty set with a binary operation is called a groupoid.

(7) $(Z-\{0\},\cdot)$ is not a group under multiplication. (false)

(8) True or false $\begin{pmatrix}1&2&3\\1&3&2\end{pmatrix}\begin{pmatrix}1&2&3\\1&3&2\end{pmatrix}=\begin{pmatrix}1&2&3\\1&2&3\end{pmatrix}$
(True)

(9) If G is an abelian group & H is subset of G Then $aH=Ha$ for some a in G.
(True)

(10) The set $(1,-1)$ is a group under (multiplication)

(11) $(Z,*)$ is a cyclic group generated by
Star, or dot (± 1) $H=\{1,-\}$

(12) A bijective mapping from non-empty set X to X is called identity permutation, A

if $ab = ba$ commutative law hold then abelian. $ab \neq ba$ — non abelian.

(25) if X has 5 elements then total no of permutations on X is $5! = 120$ $5\times4\times3\times2\times1=120$

(26) Every cyclic group is an abelian group (True)

(27) if A & B are the element of a group G then $(b^{-}a^{-1})^{-1} = ab$

(28) if X has 2 element then total no of permutation X is $2!$.

(29) The product of two even permutations & two odd permutations is an even permutation. True.

(30) The order of symmetric group (S_n, o) is n.

(31) if $f = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$ & $g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$ are two permutations then their product permutations is $fog = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 3 & 4 & 1 & 2 \end{pmatrix}$ (True)

(32) Every subsgroup of a cyclic group is cyclic True.

(33) The group $(1, -1, i, -i)$ under multiplication is a cyclic group genrated by 2 genrators $\Rightarrow (i, -i) = \pm i$

(34) The set of integers is an abelian group under addition $a*b = b*a$ $2+3 = 3+2 = 5$ $-3+2 = 2-3 = -1$

(35) The inverse permutation of $\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$.

Ans = $\begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$

(36) The order of a group is 35, which of the following cannot be the order of the subsgroup of G? 6.

$\frac{1}{2} \neq 0.5$

(37) $(\mathbb{Z}, \div)$ is (not) a groupoid under the operation of usual division. True

(38) if X has a 4 elements then total no of permutations on X is 24

(39) length of cyclic group $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 3 & 4 & 5 & 6 & 7 & 1 \end{pmatrix}$

7.

(40) Let $X = \{1, 2, 3, 4, 5\}$ if $\alpha(3) = 5$ then $\alpha^{-1}(5) = 3$

(41) which of the following is non trivial subgroup of the group $\{1, -1, i, -i\}$?

Ans = $\{1, -1\}$

(42) if X consists of the elements 1, 2, — 7 then the symbol $\{1, 3, 4, 2, 6\}$ means the permutation

Ans = $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 3 & 6 & 4 & 2 & 5 & 1 & 7 \end{pmatrix}$

(43) The group $(G, *)$ is said to be an abelian group of commutative group if for all $a, b \in G$, $a * b = b * a$.

(44) The group $\{1, -1\}$ is subgroup of the group $\{1, -1, i, -i\}$ (True).

(45)

(46)

(47)

(48)

(49)

(50)

(51)

(52)

(53)

(54)

(55)

56)

57

(58)

(45) The length of cyclic group $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 2 & 4 & 3 & 5 & 6 \end{pmatrix}$

$= (2)$ Repeated element ignore

(46) The order & index of a subgroup H of group G divide the order of the group G. (True).

(47) The order of $(1\ 2)(3\ 4)$ is 2.

(48) $(Z, +)$ is the subgroup of $(Q, \cdot)$. false

$+, \cdot$ order same nhi hai

(49) The set of n n^{th} roots of unity is a cyclic group under multiplication.

(50) Let $e \in G$ be identity element of a group G & $H \subset G$ then $e \cdot H = H$ True.

(51) if a, b are the elemants of group G then $(ab)^{-1} = b^{-1}a^{-1}$.

(52) A non empty set with the associative binary operation and an identity element is called monoid.

(53) if $fog \neq gof$ True

(54) The set $\{1, -1\}$ is group under multiplication

(55) The set $\{1, -1\}$ is a group under addition false.

(56) The group $(Z, +)$ is an abelian group.

57 what is the order of the group G $(Z, +)$ ∞.

(58) what is the order of Group G $\{1, -1, i, -i, j, -j, k, -k\}$ = 8

(58) what is the order of group $G = \{1, -1, i, -i\}$
Answer = {4}

60 The commutative law does not holds in composition of permutations ~~false~~ True
MCQ 53, 59 Same

(61) A non-empty set with a binary operation is called groupoid.

(62) The group $(z, +)$ is a finite group false
MCQ 57 & 62 Same.

(63) The set of real no is an abelian group under addition

(64) $a * H = H * a$ relation holds if H is ~~Semigroup~~ subgroup of an abelian group.

(65) The operation of adition satisfies the commutative property true.

(66) Substraction is binary operation on the set of real numbers $\mathbb{R}$. (True)

(67) A bijective mapping from non-empty set x to x is called permutations? (True)

(68) Let $H = \{1, -1, i, -i\}$ & $1 \in G$ then the right coset $H1$ is $= \{1, -1, i, -i\}$

(69) $(N, +)$ is a semigroup under the operation of usual addition. (True)

(70) Every cyclic permutation can be expressed as product of transposition

(71) if index of H in G is 4. Then there are 4 distinct left or right cosets H in G. not sure

(72) $(Z, \div)$ is Not a groupoid under the operation of usual division. True.

(73) $a*H$ is a set of left coset.

(74) The identity element of group is not unique false

(75) How many identity element are in the group $\{1, -1, i, -i\} = 1$.

(76) A group G is abelian if & only if $(ab)^2 = a^2b^2$ for all $a, b \in G$ (True)

(77) The group S_n of all permutation defined on a set X with element is called Symmetric group of degree n.

(78)

THE END

$(-1)^{-1} = 1$

$(w^2)^{-1} = w$

$2\left(\frac{1}{2}\right) = 1$

$2 - 2 = 0$

$e = 0$

$13, 15 \in N$

$13 + 15 = \notin N$

$x, y \in N, \; x + y \notin N$

$0 + x, \; x + x = x$

binary

↓

closure property holds

$x \in Z, \; -x \in Z$

circle is integer //
circle ji wi si //
zero operation

x ∘ y = 0

Binary operation :- { - ✓ ✓ hold closure property ✓ ✗ }
closure property x, y ∈ N, x+y ∈ N

Let S be a non-empty set, The binary operation * is a function

$$* : S \times S \longrightarrow S$$

which assigns each ordered pair (a, b) of S×S a unique element of S.

Suppose a, b ∈ S

a * b ∈ S

S = {1, 2, 3 ---} 2, 5 ∈ S, 2+5 = 7 ∈ S

x, y ∈ S, x+y ∈ S

lecture no #2

Groupoid: A non-empty set with a binary operation is called Groupoid.

for example, S ≠ ϕ

(S, *)

for x, y ∈ S

x * y ∈ S

so closure property holds.

e.g N = {1, 2, 3 ---}

(N, +)

3, 5 ∈ N 3+5 ∈ N

x, y ∈ N, x+y ∈ N set of N no is closed under the operation addition.

Semigroup :-

A non-empty set with an associative ② binary ① operation is called Semigroup.

e.g (1) $(S, *)$ ۔ binary ① ایک ایسا جوڑا ہے جس میں
for $x, y \in S$, $x * y \in S$
S is closed under *

(2) Associative law holds
for $x, y, z \in S$
$(x * y) * z = x * (y * z)$

Monoid : A non empty set with ② associative ① binary operation and an identity element is called monoid.

(1) Binary operation :-
Let * be the binary operation on S.
for every $x, y \in S$, $x * y \in S$
So closure property holds.

(2) Associative law hold
for $x, y, z \in S$
$(x * y) * z = x * (y * z)$

(3) Identity element :-
identity element exists in S
for all $x \in S$, $e * x = x * e = x$

∴ Addition = 0 , Multiplication = 1

Group:-

A non-empty set ② with an ① binary operation, an identity ③ element and the inverse ④ of its elements is called Group.

$S \neq \{\}$ $\{S, *\}$

(1) Closure property: for $x, y \in S$,
$x * y \in S$
So closure property holds.

(2) Associative law:-
for $x, y, z \in S$, $(x * y) * z = x * (y * z)$

(3) Identity element exists
$e * x = x = x * e$ for $x \in S$

(4) Inverse of all elements of S exists in S
for all, $x \in S$, $\exists x^{-1} \in S$ such that
$x * x^{-1} = e = x^{-1} * x$
↓
There exists

Therefore $(S, *)$ satisfied all condition. So this called a group.

Definition:-

An element a of a group G is called idempotent $a^2 = a$.

$a \in G$, $a^2 = a$

$1^2 = 1$, $0^2 = 0$

$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $I^2 = I$

Order of a group :-

The number of element in a group G is called order of the group.

$G = \{1, 2, 3\}$ $|G| = 3$

$a^n = e$ it is denoted by $O(G)$ or $|G|$

$(a)(a) = e(a)$ $(1)^1 = 1 \Rightarrow (-1)^2 = 1 \Rightarrow$

Finite group : if a group has a finite number of elements then it is called a finite group.

In simple words a finite group has a finite order.

Infinite group : if a group has a infinite number of elements, then its is called an infinite group.

In simple words, an infinite group has an infinite older.

Example: what is the order of the group $G = \{1, -1, i, -i\}$?

Since there are 4 elements in the group G so

$$O(G) = 4 \Rightarrow |G| = 4.$$

Order of element of group:

Let G be a group and $a \in G$, then the order of 'a' is the least +ve integer n such that

$$a^n = e$$

where 'e' is identity

Example:- $G = \{1, -1\}$

$e = 1$

$(1)^1 = 1$	$(-1)^4 = 1$
$(-1)^1 = -1$	$(-1)^6 = 1$
$(-1)^2 = 1$	$(-1)^8 = 1$

order of 1 is 1

order of -1 is 2 $\Rightarrow (-1)^2 = 1$

Theorem: if n is the order of an element 'a' of a group G. Then for any integer 'm'.

$a^m = e$, if and if $m = qn$ where q is an integer

$a^m = e$ if and only if n divides m.

Solution:- Given $a \in G$, $a^n = e$, n is least +ve integer let

$a^m = e$ — (1) by division algorithm, there exist $q, r \in Z$, such that

from eqn (1) & (2) $\therefore m = n$

Hence Proved $O(ab) = O(ba)$

Lecture 6

Abelian group: Let $(G, *)$ be a group

Then G is called an Abelian group or commutative group if the commutative law holds in G with respect to $*$, i.e

$$x * y = y * x \quad \text{for all } x, y \text{ in } G,$$

Non-abelian group:

Let $(G, *)$ be a group Then G is called a Non-abelian group or non-commutative group if the commutative law does not hold in G w.r.t $*$, ie

$$x * y \neq y * x \quad \text{for all } x, y \text{ in } G$$

Example: Show That The group $(Z, +)$ is an Abelian group

commutative ہے یا نہیں ثابت کرنا ہے group کو پہلے ثابت کریں کہ یہ group ہے یا نہیں اس کے لئے set کو پرکھنا ہوگا اگر set group کی تمام properties پر پورا اترے تو پھر group abelian group ہے یا نہیں یہ ثابت کرنے کے لئے commutative ثابت کرنا ہوگا

Example: Show That The set of non-singular matrices of order 2 is a Non-abelian group.

Permutation: Let X be non-empty set. A bijective mapping $\alpha : X \to X$ is called The permutation on X.

$$X \longrightarrow X$$
$$\downarrow \qquad \downarrow$$
domain co-domain

$$f = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$$

All possible permutation $(1, 2, 3)$

$3! \quad 3.2.1 = 6$

$$f_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$$

$$f_2 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$f_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$f_4 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$f_5 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$f_6 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

Identity permutation:- Let X be non-empty set, A bijective mapping $\alpha : X \to X$ is called The identity permutation on X, if $\alpha(x) = x$ for all $x \in X$

Maths Lecture 11

permutation(3) $X = \{1, 2, 3\}$, $n = 3$

Total permutation $3 \cdot 2 \cdot 1 = 6$

$$\alpha_1 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \end{pmatrix}$$

$$\alpha_2 = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 3 & 2 \end{pmatrix}$$

$$\alpha_3 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 1 & 3 \end{pmatrix}$$

$$\alpha_4 = \begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix}$$

$$\alpha_5 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 1 & 2 \end{pmatrix}$$

$$\alpha_6 = \begin{pmatrix} 1 & 2 & 3 \\ 3 & 2 & 1 \end{pmatrix}$$

Equality of permutation: if we have two permutation 'f' and 'g', $F = g$ if and only if image of every pre image in both permutation is same

$$f = \begin{pmatrix} 2 & 3 & 1 & 4 \\ 1 & 2 & 4 & 3 \end{pmatrix}$$

$$g = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}$$

$f = g$ ← second row

Inverse Permutation: Let X be a non-empty set and $\alpha : X \to X$ be a permutation on X such that

$$\alpha(x) = y \quad \forall x, y \in X$$

Then the inverse permutations $\alpha^{-1} : X \to X$ of α is defined as

$$\alpha^{-1}(y) = x \quad \forall x, y \in X$$

Let $f: X \to X$ and $g: X \to X$ be two permutations on a non empty set X, Then the permutation $fg: X \to X$ on X defined as

$$x fg = (xf) g \quad \forall x \in X$$

is called the product or composition of permutations f and g. it is also denoted by $f \circ g$.

Example: find the product of the following permutations:

$$\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 6 & 5 & 3 & 1 \end{pmatrix}, \quad \beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 4 & 1 & 2 & 3 \end{pmatrix}$$

$$\alpha\beta = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 4 & 6 & 5 & 3 & 1 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 6 & 5 & 4 & 1 & 2 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 5 & 1 & 3 & 2 & 4 & 6 \end{pmatrix}$$ Answer

$f(1) = 2 \Rightarrow g(2) = 5$
$f(2) = 4 \Rightarrow g(4) = 1$
$f(3) = 6 \Rightarrow g(6) = 3$
$f(4) = 5 = g(5) = 2$
$f(5) = 3 = g(3) = 4$
$f(6) = 1 = g(1) = 6$

find product

$$f=\begin{pmatrix}1&2&3&4\\2&1&4&3\end{pmatrix} \quad g=\begin{pmatrix}1&2&3&4\\3&2&1&4\end{pmatrix}$$

$$fog=\begin{pmatrix}1&2&3&4\\2&1&4&3\end{pmatrix}\begin{pmatrix}1&2&3&4\\3&2&1&4\end{pmatrix}$$

$$fog=\begin{pmatrix}1&2&3&4\\2&3&4&1\end{pmatrix}$$

Question two

$$\alpha=\begin{pmatrix}1&2&3&4\\2&3&4&1\end{pmatrix}, \quad \beta=\begin{pmatrix}1&2&3&4\\2&1&4&3\end{pmatrix}$$

$$\alpha\beta=\begin{pmatrix}1&2&3&4\\2&3&4&1\end{pmatrix}\begin{pmatrix}1&2&3&4\\2&1&4&3\end{pmatrix}$$

$$\alpha\beta=\begin{pmatrix}1&2&3&4\\1&4&3&2\end{pmatrix}$$

Rough work

$f(1)=2=g(2)=2$
$f(2)=1=g(1)=3$
$f(3)=4=g(4)=4$
$f(4)=3=g(3)=1$

$f(1)=2 \Rightarrow g(2)=1$
$f(2)=3 \Rightarrow g(3)=4$
$f(3)=4 \Rightarrow g(4)=3$

$f(4)=1 \Rightarrow g(1)=2$

$$f^3 = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}$$

fog, f^2og, f^3og نکالیں اور f^4 بھی find کریں

$$\begin{pmatrix} 1 & 2 & 3 & 4 \\ 4 & 1 & 2 & 3 \end{pmatrix}$$

Lecture no 15 :-

Cyclic Permutations: Let $X = \{x_1, x_2, x_3 \cdots x_n\}$

Then a permutation α is called cyclic Permutation or cycle of length n if

$$\alpha = \begin{pmatrix} x_1 & x_2 & x_3 \cdots & x_n \\ x_2 & x_3 & x_4 \cdots & x_1 \end{pmatrix}$$

$$\text{cycle} = (x_1, x_2\ x_3 \cdots x_n)$$

e.g $\begin{pmatrix} 1 & 2 & 3 \\ 2 & 3 & 1 \end{pmatrix} = (1\ 2\ 3)$

e.g $\begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix} = (1\ 2\ 3\ 4)$

e.g $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 2 & 3 & 4 & 1 & 5 \end{pmatrix} = (1\ 2\ 3\ 4)(5)$ ignore

$= (1\ 2\ 3\ 4)$

Example: find the following product & express it as a product of cyclic permutations on mutually disjoint sets.

$$(14)(235)(35)(45)$$

Solution:

cycle form $(1\ 4)(2\ 3\ 5)(3\ 5)(4\ 5)$

$$= \begin{pmatrix}1 & 4\\ 4 & 1\end{pmatrix}\begin{pmatrix}2 & 3 & 5\\ 3 & 5 & 2\end{pmatrix}\begin{pmatrix}3 & 5\\ 5 & 3\end{pmatrix}\begin{pmatrix}4 & 5\\ 5 & 4\end{pmatrix}$$

$$= \begin{pmatrix}1 & 2 & 3 & 4 & 5\\ 4 & 2 & 3 & 1 & 5\end{pmatrix}\begin{pmatrix}1 & 2 & 3 & 4 & 5\\ 1 & 3 & 5 & 4 & 2\end{pmatrix}\begin{pmatrix}1 & 2 & 3 & 4 & 5\\ 1 & 2 & 5 & 4 & 3\end{pmatrix}\begin{pmatrix}1 & 2 & 3 & 4 & 5\\ 1 & 2 & 3 & 5 & 4\end{pmatrix}$$

Then find Product

$$= \begin{pmatrix}1 & 2 & 3 & 4 & 5\\ 5 & 4 & 3 & 1 & 2\end{pmatrix} \Rightarrow \begin{pmatrix}1 & 5 & 2 & 4\\ 5 & 2 & 4 & 1\end{pmatrix}$$

$= (1\ 2\ 3\ 4\ 5)$

$= (1\ 5\ 2\ 4) \rightarrow$ disjoint form.

length of cycle = 4

$$= \begin{pmatrix} 1 & 8 \\ 8 & 1 \end{pmatrix} \begin{pmatrix} 3 & 6 & 4 \\ 6 & 4 & 3 \end{pmatrix} \begin{pmatrix} 5 & 7 \\ 7 & 5 \end{pmatrix}$$

$= (1\ 8)(3\ 6\ 4)(5\ 7)$

disjoint cycle.

Lecture no 16 :-

The length of the cycle permutation $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 3 & 4 & 5 & 6 & 7 & 1 \end{pmatrix}$ is - - -

Sol:

$$= \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 \\ 2 & 3 & 4 & 5 & 6 & 7 & 1 \end{pmatrix}$$

$= (1\ 2\ 3\ 4\ 5\ 6\ 7)$

length of cycle is = 7

Lecture no 17 :

theorem pdf sy.

Symmetric group :-

The group S_n of all permutations defined on a set X with n elements is called symmetric group of degree n.

$$\begin{pmatrix} 1 & 2 & 3 & 4 & 5 \\ 5 & 3 & 2 & 4 & 1 \end{pmatrix}$$

Sol: $= \begin{pmatrix} 1 & 5 \\ 5 & 1 \end{pmatrix}\begin{pmatrix} 4 \\ 4 \end{pmatrix}\begin{pmatrix} 2 & 3 \\ 3 & 2 \end{pmatrix}$

$= (1\ 5)(2\ 3)$

Answer

Lecture 19:

Even Permutation: A permutation α in S_n is called an even permutation if it can be written as a product of even number of transposition.

Example $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 4 & 6 & 1 & 3 & 5 & 8 & 9 & 7 \end{pmatrix}$

$= \begin{pmatrix} 1 & 2 & 4 \\ 2 & 4 & 1 \end{pmatrix}\begin{pmatrix} 3 & 6 & 5 \\ 6 & 5 & 3 \end{pmatrix}\begin{pmatrix} 7 & 8 & 9 \\ 8 & 9 & 7 \end{pmatrix}$

$= (1\ 2\ 4)(3\ 6\ 5)(7\ 8\ 9)$

$= (1\ 2)(1\ 4)(3\ 6)(3\ 5)(7\ 8)(7\ 9)$

Therefore no of transposition is $= 6$ So the Permutation is even.

odd permutation

A permutation α in S_n is called an even permutation if it can be written as a product of even number of transposition.

Example: $\begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ 2 & 4 & 6 & 1 & 3 & 5 & 8 & 9 & 7 \end{pmatrix}$

$= (1\ 2\ 4)$ $= \begin{pmatrix} 1 & 2 & 4 \\ 2 & 4 & 1 \end{pmatrix}\begin{pmatrix} 3 & 6 & 5 \\ 6 & 5 & 3 \end{pmatrix}\begin{pmatrix} 7 & 8 & 9 \\ 8 & 9 & 7 \end{pmatrix}$

$= (1\ 2\ 4)(3\ 6\ 5)(7\ 8\ 9)$

$= (1\ 2)(1\ 4)(3\ 6)(3\ 5)(7\ 8)(7\ 9)$

Therefore no of transposition is 9. So This is odd Permutation.

Lecture 20 The permutation $(1\ 2)(1\ 3)$ is even permutation.

True ✓

False